

Phase 10A - limitations

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PHASE 10A — LIMITATIONS

Consolidated from the phase report, per Master Prompt Section 16.

1. No genuine .pbix file exists in this deliverable

This build environment has no Power BI Desktop and cannot produce a .pbix binary. What is delivered instead — governed data marts (`data_marts/*.csv`), data-accurate HTML dashboard mockups (`dashboards/*.html`), and a literal build guide with a DAX measures library (`docs/power_bi_architecture.md`) — is the honest substitute: everything a Power BI developer needs to produce the real file, reviewable by stakeholders before that work happens.

2. All 7 dashboards run on the n=26 demo batch, not the full 8,000-property portfolio

Every prior phase (6, 7, 9) built and tested its outputs against the same 25-property demo sample (properties with full expense+rental history) plus property 1 (added specifically to demonstrate the INSUFFICIENT path). These dashboards inherit that same sample rather than the full portfolio, for the same reason: full-portfolio financial/risk/decision evaluation was never run in any prior phase, so there is no governed full-portfolio output to source from. Dashboard 1's and Dashboard 5's KPIs that reference the full 8,000-property / 90-locality counts (from `properties.csv` / `localities.csv` directly) are the exception — those are simple counts, not evaluated outputs.

3. Dashboard rendering could not be fully screenshot-verified in this sandbox

The HTML dashboards load Chart.js and Google Fonts from CDNs (`cdnjs.cloudflare.com` , `fonts.googleapis.com`) for interactivity and typography. This sandbox's network egress allowlist does not include those domains, so this phase's own QA screenshots (via `wkhtmltoimage`) show the correct page structure, layout, tables, and KPI values, but not the actual charts or web font — those will render correctly in the person's own browser, which has normal internet access, but that was not directly observable here. The one real bug this QA process *did* catch (Section 5 of the phase report — property 1's missing locality name) was visible even without the charts rendering, which is what made it catchable at all.

4. The DAX measures library is documentation, not tested code

`docs/power_bi_architecture.md` Section 4 lists DAX formulas written for a real Power BI data model. No Power BI runtime exists in this environment to execute or syntax-check them. A developer implementing the real report should treat these as a starting reference, not copy-paste-and-trust code.

5. Refresh automation is a documented target only

Section 7 of the architecture doc lists a suggested refresh cadence per mart. No scheduling, pipeline, or automation exists in this deliverable to actually implement it.

6. Governance status columns must not be dropped by whoever builds the real Power BI report

Every mart preserves the exact status vocabulary from its source phase (`PASS` , `PASS WITH LIMITATIONS` , `CONDITIONAL` , `REJECTED` , `UNCALIBRATED` , `INSUFFICIENT DATA`). It would be easy, and wrong, for a Power BI developer optimizing for a “cleaner-looking” dashboard to collapse these into a single green/red indicator — doing so would erase exactly the honesty this platform has built through Phases 1-9. This is called out explicitly in the architecture doc (Section 6) so it isn’t lost in translation to the real tool.

7. Dashboard 2’s locality-level YoY/MoM growth is computed fresh in this phase, not copied from Phase 8

Phase 8 computed a locality-level appreciation CAGR (for the financial engine’s use in Phase 6/7), but not a YoY/MoM price-growth figure formatted for a market dashboard. `build_market_locality_mart()` computes YoY/MoM directly from `core.market_monthly` — the same source table, same underlying values, different aggregation window than Phase 6/8’s CAGR — so a reader comparing this dashboard’s YoY growth figure to Phase 6’s appreciation-rate assumption for the same locality should expect them to differ (different time windows, different purposes), not treat a mismatch as an error.

8. Post-delivery audit fix: inherited the Phase 6 ROI bug propagated further than in Phase 7/9 (now fixed)

`src/finance/engine.py` , and the static `sources/phase6/*.csv` / `sources/phase7/05_extended_scenario_comparison.csv` files this phase’s mart builder reads, were all pulled in *before* Phase 6’s own post-delivery audit fixed a bug in `roi_total_holding_period` (it was computing a gross cash multiple instead of a net return, overstating ROI by exactly 100 percentage points in every case – the same inherited-stale-copy issue independently found in Phases 7 and 9). Unlike those two phases, here the bug reached the actual Power BI-facing data layer: `mart_04_financial_risk.csv` and `mart_04c_scenarios_extended4.csv` carried the wrong values (though no dashboard HTML visibly labels `roi_total_holding_period` as a rendered number – only cap rate, IRR, NPV, DSCR, and Monte Carlo probabilities are shown). **Fixed** by re-copying Phase 6’s corrected `engine.py` and the four affected source CSVs (property_id sets verified identical before substituting), then rerunning `build_marts.py` and all 7 dashboard-generation scripts. Portfolio KPIs and the decision distribution were independently re-derived from the regenerated marts and came back identical to what was already reported, confirming no decision-relevant figure was ever affected.

9. No automated test suite existed for this phase prior to audit (now added)

`src/marts/test_phase10a.py` (9 tests) was added during the same review – this phase had inherited `test_phase6.py` / `test_phase8.py` from its source phases but had nothing testing the marts or dashboards it actually builds. All 9 passing; see Section 5B of `docs/phase10a_report.md` for what each test checks.